Q.30 What is need of transistor biasing. (CO5)

Q.31 How N and P type semiconductors are formed (CO4)

Q.32 Explain with diagram the working of JFET as an amplifier? (CO8)

Q.33 Explain the characteristics of ideal diode. (CO3)

Q.34 Write a short note on Filter circuit (CO6)

Q.35 What are the advantages of multistage amplifiers? (CO7)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Write short note on any two of the following:- (Co8)

a) LED b) BJT

c) MOSFET

Q.37 Difference between CB, CC and CE configuration. (CO4)

Q.38 Explain the working of Single Stage Amplifier with its block diagram and its physical and graphical explanation. (Co7)

No. of Printed Pages : 4 120933/030933

Roll No. ....

3rd Sem / Electrical, GE, Power Station Engg., Elect. & Eltx. Engg., Fire Tech & Safety

Subject:- Electronics-I / Basic Electronics

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The majority carriers in P-type semiconductor are (CO2)

a) holes

b) electrons

c) both holes and electrons

d) none of the above

Q.2 N-type semiconductor is formed by adding: (CO4)

a) Trivalent b) Pentavalent

c) Quadrant d) None

Q.3 Semiconductors have_____ bonds (CO3)

a) Ionic b) Covalent

c) Both a & b d) None

Q.4 Zener diode is used as (CO3)

a) Amplifier b) Voltage Regulator

c) Coupler d) Rectifier

Q.5 The emitter is_______ doped (CO4)

a) Lightly b) Heavily

c) Moderately d) None

Q.6 The forbidden energy gap in semiconductorslies: (CO3)

a) Below the valence band
b) above the conduction band
c) between the conduction band and valence band
d) is the same as valence band

Q.7 The transistor works as amplifier in______ region. (CO4)

a) Saturation b) cut off
c) Inverted d) active

Q.8 The function of emitter in NPN-transistor is (CO4)

a) to emit or inject holes into collector
b) to emit or inject electrons into collector
c) to emit or inject electrons into base
d) to emit or inject holes in base

Q.9 Photodiode is designed to work in ______ region. (CO3)

a) forward bias
b) reverse bias
c) both forward and reverse bias
d) None of the above

Q.10 An field effect transistor is essentially a______ (CO8)

a) current driven device
b) voltage driven device
c) power driven device
d) None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What is passive component? (CO1)

Q.12 BJT stands for_______. (CO4)

Q.13 LED stands for________. (CO3)

Q.14 Rectifiers are used to convert DC into AC. (True/False) (CO3)

Q.15 Define intrinsic semiconductor? (CO2)

Q.16 Define doping? (CO2)

Q.17 What is JFET? (CO8)

Q.18 Name any two passive components. (CO1)

Q.19 Capacitor is a passive component. (True/False) (CO1)

Q.20 Define Diffusion current. (CO2)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain forward and reverse biasing of diode. (CO3)

Q.22 Explain the working of NPN transistor. (CO4)

Q.23 Explain the working of PNP transistor. (CO4)

Q.24 Explain drift and diffusion current. (CO2)

Q.25 What is zener diode and its characteristics. (CO3)

Q.26 Explain half wave rectifier and its efficiency. (CO3)

Q.27 Explain Full wave rectifier. (CO3)

Q.28 Discuss the atomic structure of N-typesemiconductors (CO2)

Q.29 Write a note on Schottkey diode (CO3)